Unit: Unit 7: Factoring
Topic: Sum and Difference of Cubes

Name: ____________________
Date: ____________

Advanced (Habanero)

Q1. What is the factored form of $x^3 + 8$?
(A) $(x+2)(x^2-2x+4)$
(B) $(x-2)(x^2+2x+4)$
(C) $(x+2)(x^2+2x+4)$
(D) $(x-2)(x^2-2x+4)$
(E) $(x+2)(x^2-2x-4)$

Q2. Factor the expression $27x^3 - 8y^3$.
(A) $(3x-2y)(9x^2+6xy+4y^2)$
(B) $(3x+2y)(9x^2-6xy+4y^2)$
(C) $(3x-2y)(9x^2-6xy+4y^2)$
(D) $(3x+2y)(9x^2+6xy+4y^2)$
(E) $(3x-2y)(9x^2+6xy-4y^2)$

Q3. What is the factored form of $64 - x^3$?
(A) $(4-x)(16+4x+x^2)$
(B) $(4+x)(16-4x+x^2)$
(C) $(4-x)(16-4x+x^2)$
(D) $(4+x)(16+4x+x^2)$
(E) $(4-x)(16+4x-x^2)$

Q4. Factor the expression $x^3 + 64$.
(A) $(x+4)(x^2-4x+16)$
(B) $(x-4)(x^2+4x+16)$
(C) $(x+4)(x^2+4x+16)$
(D) $(x-4)(x^2-4x+16)$
(E) $(x+4)(x^2-4x-16)$

Q5. What is the factored form of $8x^3 - 27$?
(A) $(2x-3)(4x^2+6x+9)$
(B) $(2x+3)(4x^2-6x+9)$
(C) $(2x-3)(4x^2-6x+9)$
(D) $(2x+3)(4x^2+6x+9)$
(E) $(2x-3)(4x^2+6x-9)$

Q6. Factor the expression $125x^3 - 8y^3$.
(A) $(5x-2y)(25x^2+10xy+4y^2)$
(B) $(5x+2y)(25x^2-10xy+4y^2)$
(C) $(5x-2y)(25x^2-10xy+4y^2)$
(D) $(5x+2y)(25x^2+10xy+4y^2)$
(E) $(5x-2y)(25x^2+10xy-4y^2)$

Q7. What is the factored form of $x^3 - 125$?
(A) $(x-5)(x^2+5x+25)$
(B) $(x+5)(x^2-5x+25)$
(C) $(x-5)(x^2-5x+25)$
(D) $(x+5)(x^2+5x+25)$
(E) $(x-5)(x^2+5x-25)$

Q8. Factor the expression $216 - x^3$.
(A) $(6-x)(36+6x+x^2)$
(B) $(6+x)(36-6x+x^2)$
(C) $(6-x)(36-6x+x^2)$
(D) $(6+x)(36+6x+x^2)$
(E) $(6-x)(36+6x-x^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^3 + 8y^3$ and show all steps.

Q10. Factor the expression $27x^3 - y^3$ and identify the pattern.

Chef's Tips

- Remind students to review the difference of cubes formula.
- Encourage students to check their work by multiplying the factors.
- Suggest that students use the sum and difference of cubes formulas to factor expressions.